

Algebra and Discrete Mathematics (ADM)

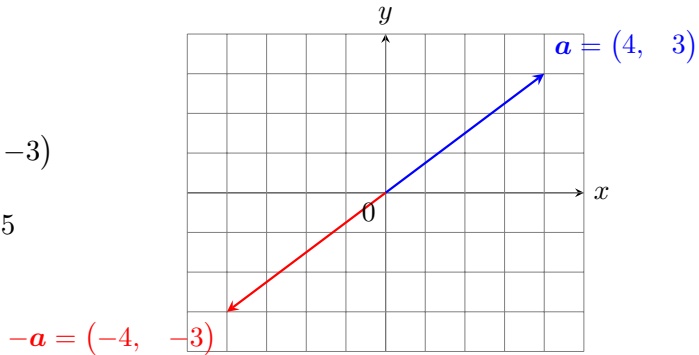
Tutorial 1 Vectors and matrices

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Vector

$$\mathbf{a} = (4, 3), \quad -\mathbf{a} = (-4, -3)$$

$$\|\mathbf{a}\| = \sqrt{4^2 + 3^2} = \sqrt{16 + 9} = 5$$



Vector addition and subtraction

$$\mathbf{a} = (1, -3, 2, 5), \mathbf{b} = (2, 2, 4, 0)$$

$$\mathbf{a} + \mathbf{b} = ?$$

$$\mathbf{a} - \mathbf{b} = ?$$

$$\mathbf{b} - \mathbf{a} = ?$$

Vector addition and subtraction

$$\mathbf{a} = (1, -3, 2, 5), \mathbf{b} = (2, 2, 4, 0)$$

$$\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a} = (1 + 2, -3 + 2, 2 + 4, 5 + 0) = (3, -1, 6, 5)$$

$$\mathbf{a} - \mathbf{b} = (1 - 2, -3 - 2, 2 - 4, 5 - 0) = (-1, -5, -2, 5)$$

$$\mathbf{b} - \mathbf{a} = (1, 5, 2, -5) = -(\mathbf{a} - \mathbf{b})$$

Projection vectors

- Let $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ with $\mathbf{a} \neq \mathbf{0}$ and $\mathbf{b} \neq \mathbf{0}$.
- Projection of $\mathbf{a}$ onto $\mathbf{b}$ is given by

$$\text{proj}_{\mathbf{b}} \mathbf{a} = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{b}\|^2} \mathbf{b}$$

- Projection of $\mathbf{b}$ onto $\mathbf{a}$ is given by

$$\text{proj}_{\mathbf{a}} \mathbf{b} = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|^2} \mathbf{a}$$

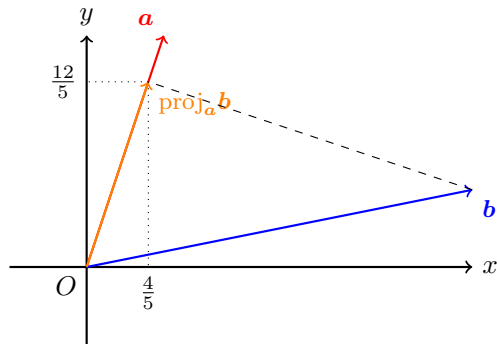
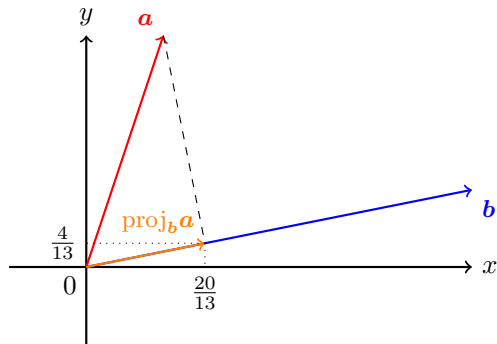
Example

$$\mathbf{a} = (1, 3), \mathbf{b} = (5, 1)$$

$$\text{proj}_{\mathbf{b}} \mathbf{a} = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{b}\|^2} \mathbf{b} = \frac{1 \times 5 + 3 \times 1}{5^2 + 1} \mathbf{b} = \frac{8}{26} (5, 1) = \left(\frac{20}{13}, \frac{4}{13} \right)$$

$$\text{proj}_{\mathbf{a}} \mathbf{b} = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|^2} \mathbf{a} = \frac{1 \times 5 + 3 \times 1}{1^2 + 3^2} \mathbf{a} = \frac{8}{10} (1, 3) = \left(\frac{4}{5}, \frac{12}{5} \right)$$

Projection vectors



$$\mathbf{a} = (1, 3), \quad \mathbf{b} = (5, 1), \quad \text{proj}_{\mathbf{b}} \mathbf{a} = \left(\frac{20}{13}, \frac{4}{13} \right), \quad \text{proj}_{\mathbf{a}} \mathbf{b} = \left(\frac{4}{5}, \frac{12}{5} \right)$$

Projection theorem

Theorem

Let $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ with $\mathbf{a} \neq \mathbf{0}$. Then $\mathbf{b}$ can be uniquely written as $\mathbf{b} = \mathbf{w}_1 + \mathbf{w}_2$, where $\mathbf{w}_1$ is a scalar multiple of $\mathbf{a}$ and $\mathbf{w}_2$ is orthogonal to $\mathbf{a}$.

Proof.

$$\mathbf{b} = \mathbf{w}_1 + \mathbf{w}_2 = \alpha \mathbf{a} + \mathbf{w}_2$$

Then

$$\mathbf{b} \cdot \mathbf{a} = (\alpha \mathbf{a} + \mathbf{w}_2) \cdot \mathbf{a} = \alpha \|\mathbf{a}\|^2 + (\mathbf{w}_2 \cdot \mathbf{a}) = \alpha \|\mathbf{a}\|^2 \implies \alpha = \frac{\mathbf{b} \cdot \mathbf{a}}{\|\mathbf{a}\|^2}$$

is the only possible value for α . Define

$$\mathbf{w}_1 = \frac{\mathbf{b} \cdot \mathbf{a}}{\|\mathbf{a}\|^2} \mathbf{a} = \text{proj}_{\mathbf{a}} \mathbf{b}, \quad \mathbf{w}_2 = \mathbf{b} - \mathbf{w}_1.$$

Then $\mathbf{b} = \mathbf{w}_1 + \mathbf{w}_2$, and $\mathbf{w}_2 \cdot \mathbf{a} = 0$.



Projection theorem

- $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$, with $\mathbf{a} \neq \mathbf{0}$

$$\mathbf{b} = \text{proj}_{\mathbf{a}} \mathbf{b} + \mathbf{w}_2$$

- $\text{proj}_{\mathbf{a}} \mathbf{b}$ is called the *component of $\mathbf{b}$ along $\mathbf{a}$*
- $\mathbf{b} - \text{proj}_{\mathbf{a}} \mathbf{b}$ is called the *component of $\mathbf{b}$ orthogonal to $\mathbf{a}$*

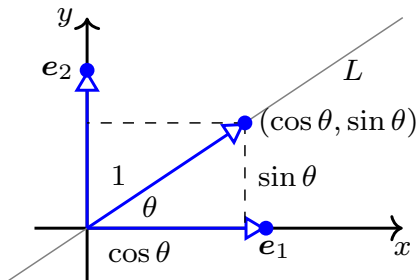
Orthogonal projection on a line

- Find the orthogonal projections of the vectors $e_1 = (1, 0)$ and $e_2 = (0, 1)$ on the line L through the origin that makes an angle θ with the positive x -axis in $\mathbb{R}^2$
- First we find the orthogonal projection of e_1 onto $a := (\cos \theta, \sin \theta)$

$$\begin{aligned}\text{proj}_a e_1 &= \frac{e_1 \cdot a}{\|a\|^2} a = \frac{\cos \theta + 0}{1} (\cos \theta, \sin \theta) \\ &= (\cos^2 \theta, \sin \theta \cos \theta)\end{aligned}$$

- For any nonzero vector u on the line L , we have $u = \alpha a$ for some $\alpha \in \mathbb{R} \setminus \{0\}$

$$\text{proj}_u e_1 = \frac{\alpha(e_1 \cdot a)}{\alpha^2 \|a\|^2} (\alpha a) = \frac{e_1 \cdot a}{\|a\|^2} a$$



- Similarly

$$\begin{aligned}\text{proj}_u e_2 &= \frac{e_2 \cdot a}{\|a\|^2} a \\ &= (\sin \theta \cos \theta, \sin^2 \theta)\end{aligned}$$

Matrices

$$A = [1] \in \mathcal{M}_{1 \times 1}, \quad B = (1, \ 2) \in \mathcal{M}_{1 \times 2}, \quad C = \begin{pmatrix} 3 \\ 5 \end{pmatrix} \in \mathcal{M}_{2 \times 1}, \quad D = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \in \mathcal{M}_{2 \times 2}$$

$$E = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \in \mathcal{M}_{2 \times 3}$$

Special matrices

upper triangular matrix $A = \begin{pmatrix} 1 & 1 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & -1 \end{pmatrix}$

lower triangular matrix $B = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ 1 & 0 & -3 \end{pmatrix}$

diagonal matrix $C = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$, $D = \begin{pmatrix} d_{11} & 0 & \dots & 0 \\ 0 & d_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & d_{nn} \end{pmatrix}$

zero matrix $O = \begin{pmatrix} 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{pmatrix}$

Transpose of a matrix

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}, \quad A^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$

Matrix addition

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}, \quad B = \begin{pmatrix} -1 & 0 & 2 \\ 3 & 5 & 1 \\ -2 & 2 & -3 \end{pmatrix}$$

$$A + B = B + A = \begin{pmatrix} 1 + (-1) & 2 + 0 & 3 + 2 \\ 4 + 3 & 5 + 5 & 6 + 1 \\ 7 + (-2) & 8 + 2 & 9 + (-3) \end{pmatrix} = \begin{pmatrix} 0 & 2 & 5 \\ 7 & 10 & 7 \\ 5 & 10 & 6 \end{pmatrix}$$

Matrix subtraction

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}, \quad B = \begin{pmatrix} -1 & 0 & 2 \\ 3 & 5 & 1 \\ -2 & 2 & -3 \end{pmatrix}$$

$$A - B = -(B - A) = \begin{pmatrix} 1 - (-1) & 2 - 0 & 3 - 2 \\ 4 - 3 & 5 - 5 & 6 - 1 \\ 7 - (-2) & 8 - 2 & 9 - (-3) \end{pmatrix} = \begin{pmatrix} 2 & 2 & 1 \\ 1 & 0 & 5 \\ 9 & 6 & 12 \end{pmatrix}$$

Matrix multiplication

$$A \in \mathcal{M}_{m \times n}, B \in \mathcal{M}_{n \times r}, C = AB \in \mathcal{M}_{m \times r}$$

$$A = \begin{pmatrix} 1 & 1 \\ 2 & 2 \\ 3 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 & -2 \\ -4 & 5 & 6 \end{pmatrix}$$

$$AB = \begin{pmatrix} 1 \times 1 + 1 \times (-4) & 1 \times 0 + 1 \times 5 & 1 \times (-2) + 1 \times 6 \\ 2 \times 1 + 2 \times (-4) & 2 \times 0 + 2 \times 5 & 2 \times (-2) + 2 \times 6 \\ 3 \times 1 + 3 \times (-4) & 3 \times 0 + 3 \times 5 & 3 \times (-2) + 3 \times 6 \end{pmatrix} = \begin{pmatrix} -3 & 5 & 4 \\ -6 & 10 & 8 \\ -9 & 15 & 12 \end{pmatrix}$$

$$BA = \begin{pmatrix} 1 \times 1 + 0 \times 2 + (-2) \times 3 & 1 \times 1 + 0 \times 2 + (-2) \times 3 \\ -4 \times 1 + 5 \times 2 + 6 \times 3 & -4 \times 1 + 5 \times 2 + 6 \times 3 \end{pmatrix} = \begin{pmatrix} -5 & -5 \\ 24 & 24 \end{pmatrix}$$

Matrix multiplication

$$C = \begin{pmatrix} 1 & -1 \\ 5 & 3 \end{pmatrix}$$

$$C^2 = \begin{pmatrix} 1 & -1 \\ 5 & 3 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 5 & 3 \end{pmatrix} = \begin{pmatrix} 1 \times 1 + (-1) \times 5 & 1 \times (-1) + (-1) \times 3 \\ 5 \times 1 + 3 \times 5 & 5 \times (-1) + 3 \times 3 \end{pmatrix} = \begin{pmatrix} -4 & -4 \\ 20 & 4 \end{pmatrix}$$

$$CI_2 = \begin{pmatrix} 1 & -1 \\ 5 & 3 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 \times 1 + (-1) \times 0 & 1 \times 0 + (-1) \times 1 \\ 5 \times 1 + 3 \times 0 & 5 \times 0 + 3 \times 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 5 & 3 \end{pmatrix}$$

$$I_2C = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 5 & 3 \end{pmatrix} = \begin{pmatrix} 1 \times 1 + 0 \times 5 & 1 \times (-1) + 0 \times 3 \\ 0 \times 1 + 1 \times 5 & 0 \times (-1) + 1 \times 3 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 5 & 3 \end{pmatrix}$$

Matrix multiplication

$$A = \begin{pmatrix} 1 & 1 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 3 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & 5 & 2 \end{pmatrix}, \quad AB \neq BA$$

$$\begin{aligned} AB &= \begin{pmatrix} 1 \times 3 + 1 \times (-3) + (-1) \times 4 & 1 \times 0 + 1 \times 1 + (-1) \times 5 & 1 \times 0 + 1 \times 0 + (-1) \times 2 \\ 0 \times 3 + 1 \times (-3) + 2 \times 4 & 0 \times 0 + 1 \times 1 + 2 \times 5 & 0 \times 0 + 1 \times 0 + 2 \times 2 \\ 0 \times 3 + 0 \times (-3) + 1 \times 4 & 0 \times 0 + 0 \times 1 + 1 \times 5 & 0 \times 0 + 0 \times 0 + 1 \times 2 \end{pmatrix} \\ &= \begin{pmatrix} -4 & -4 & -2 \\ 5 & 11 & 4 \\ 4 & 5 & 2 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} BA &= \begin{pmatrix} 3 \times 1 + 0 \times 0 + 0 \times 0 & 3 \times 1 + 0 \times 1 + 0 \times 0 & 3 \times (-1) + 0 \times 2 + 0 \times 1 \\ -3 \times 1 + 1 \times 0 + 0 \times 0 & -3 \times 1 + 1 \times 1 + 0 \times 0 & -3 \times (-1) + 1 \times 2 + 0 \times 1 \\ 4 \times 1 + 5 \times 0 + 2 \times 0 & 4 \times 1 + 5 \times 1 + 2 \times 0 & 4 \times (-1) + 5 \times 2 + 2 \times 1 \end{pmatrix} \\ &= \begin{pmatrix} 3 & 3 & -3 \\ -3 & -2 & 5 \\ 4 & 9 & 8 \end{pmatrix} \end{aligned}$$